Project Progress Report

Project-I (PROJ-CS-302A)

Date:

Project Title: EduSchedular: Intelligent Academic Schedular

Project Group No: G52

Supervisor's Name: Dr Stuti Mehla

Name of project lab teacher: Ms Deepti Dhingra

Team Members:

Sr. No	Roll No.	Name	Project Lab Attendance (%)
1.	2823205	Ankit kumar Sah	
2.	2323248	Vipul Gupta	
3.	2823203	Yash	
4.	2823194	Golu gupta	

Functionalities of the System Completed:

The developed system is a comprehensive Academic ERP and Communication Platform that integrates scheduling, management, collaboration, and academic evaluation functionalities.

1. Authentication & Authorization

   - Secure login using JWT-based authentication
   - Role-Based Access Control (Admin, Teacher, Student)
   - Google OAuth integration with Calendar synchronization

2. Academic & User Management

   - User management for Students, Teachers, and Admins
   - CRUD operations for Departments, Courses, Sections, and Classrooms
   - Faculty-course allocation and academic structure management

3. Timetable & Scheduling

   - Centralized timetable creation and management
   - Personalized timetable views for teachers and students
   - True swap functionality for conflict resolution
   - Visual indicators for swapped classes
   - Real-time classroom availability tracking

4. Meeting & Communication System

   - On-demand meeting creation with Google Meet integration
   - Role-based meeting broadcasting (All, Teachers, Students, Department, Section)
   - Real-time notification system with targeted delivery
   - File-based communication (PDF, Word, Excel, Images) via notifications

5. Faculty Management

    - Workload analytics and tracking
    - Leave request and approval system
    - Class swap request and approval mechanism

6. Student Academic Tracking

    - Attendance management system
    - Assessment creation and marks entry by teachers
    - Bulk marks upload support
    - Student performance analytics dashboards

7. Assignment & Content Sharing

    - Assignment creation with file upload support (PDF/Image/Text)
    - Distribution of study materials via notification system
    - Role-based access to academic resources

8. Resource Management

    - Booking system for classrooms, labs, and equipment
    - Conflict detection and availability tracking

9. Bulk Data Management

    - CSV/Excel-based bulk import for:
        - Students and Teachers
        - Courses, Departments, Sections, and Rooms
    - Validation and error reporting for uploaded data

10. User Interface Enhancements

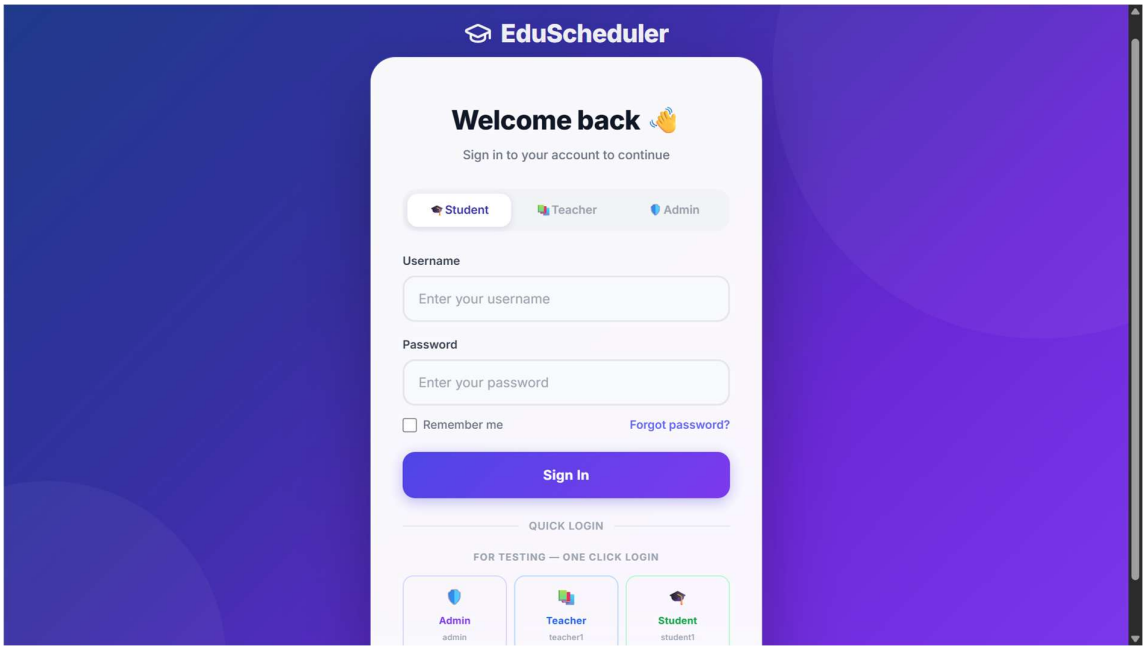
    - Table-based list views for all management pages
    - Advanced filtering (Department, Year, Section)
    - Search, sorting, and improved data visualization
    - Responsive and user-friendly dashboard design

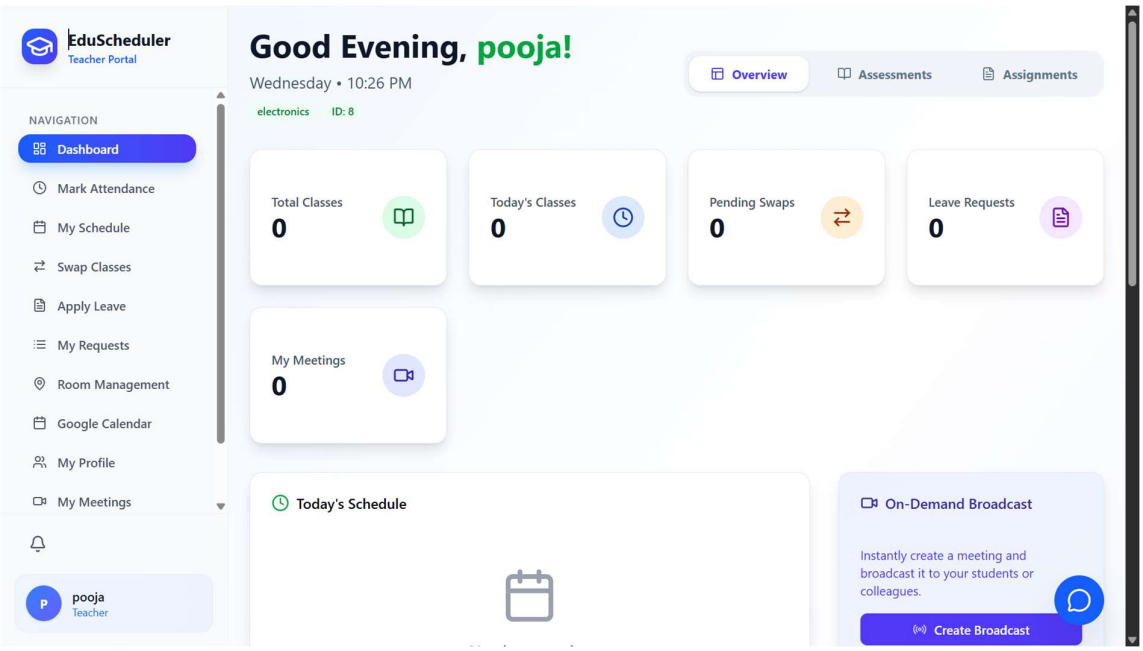
11. AI Assistance

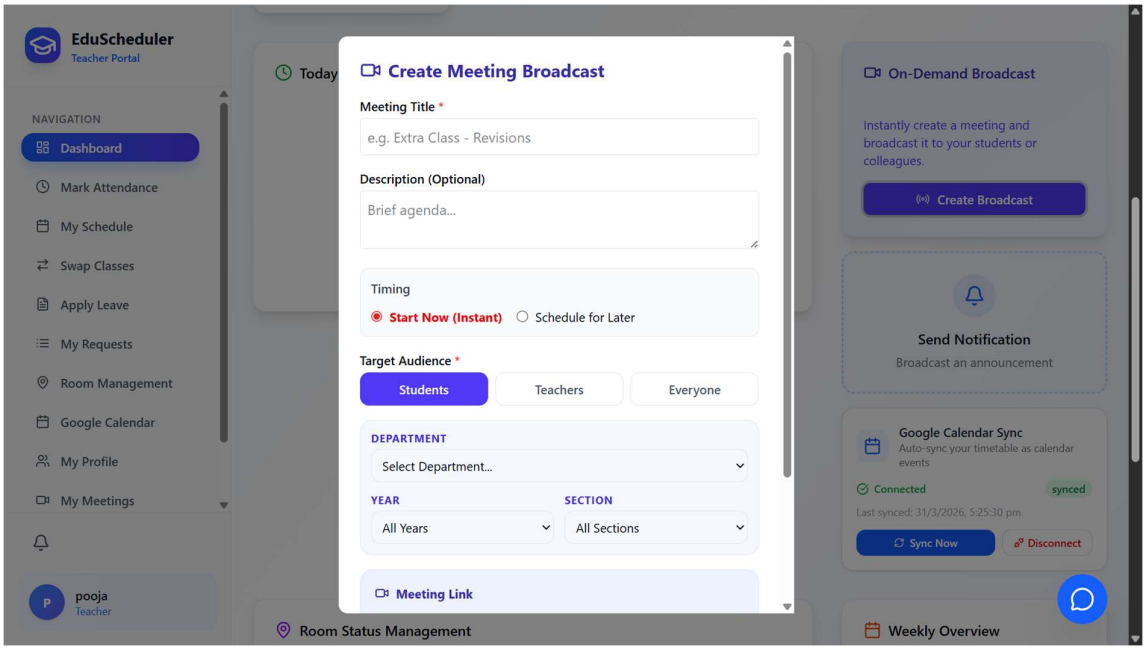
    - Integrated AI chatbot for user support and query resolution

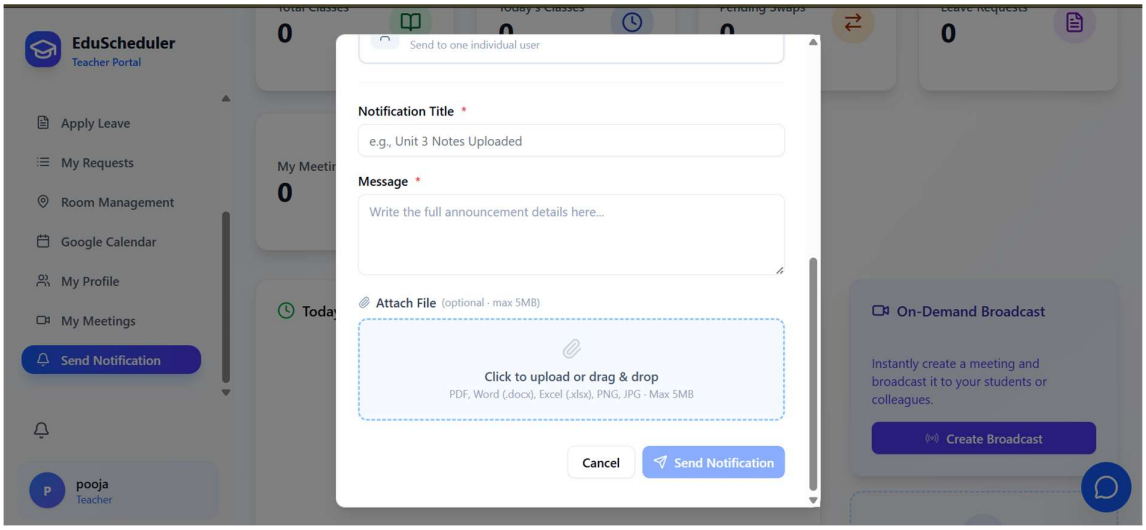
Final Summary:

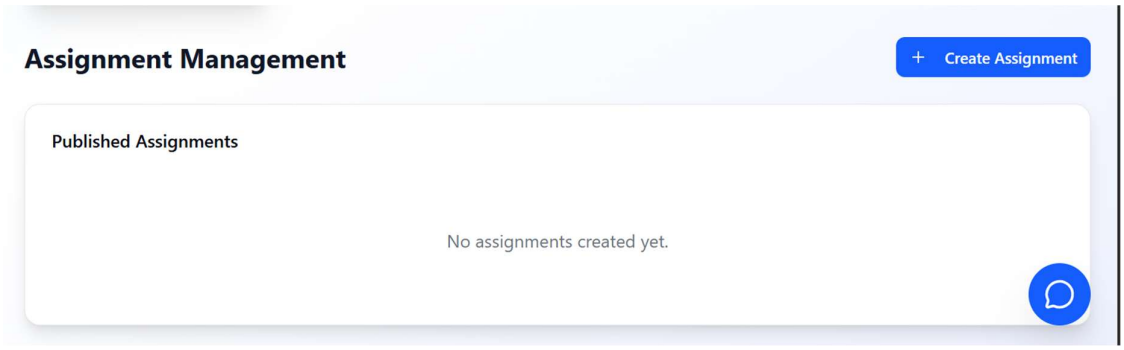
The system has evolved into a full-scale Academic ERP with integrated communication and learning management capabilities, enabling efficient scheduling, real-time collaboration, academic tracking, and scalable data management within a unified platform.

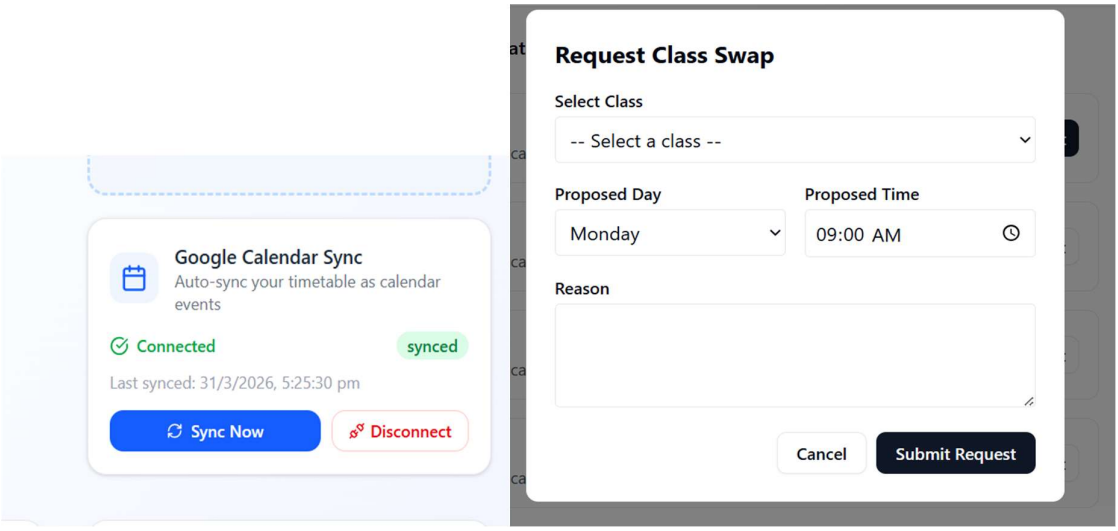

EduScheduler
Welcome back
Sign in to your account to continue
Student
Teacher
Admin
Username
Enter your username
Password
Enter your password
Remember me
Forgot password?
Sign In
QUICK LOGIN
FOR TESTING — ONE CLICK LOGIN
Admin
admin
Teacher
teacher1
Student
student1

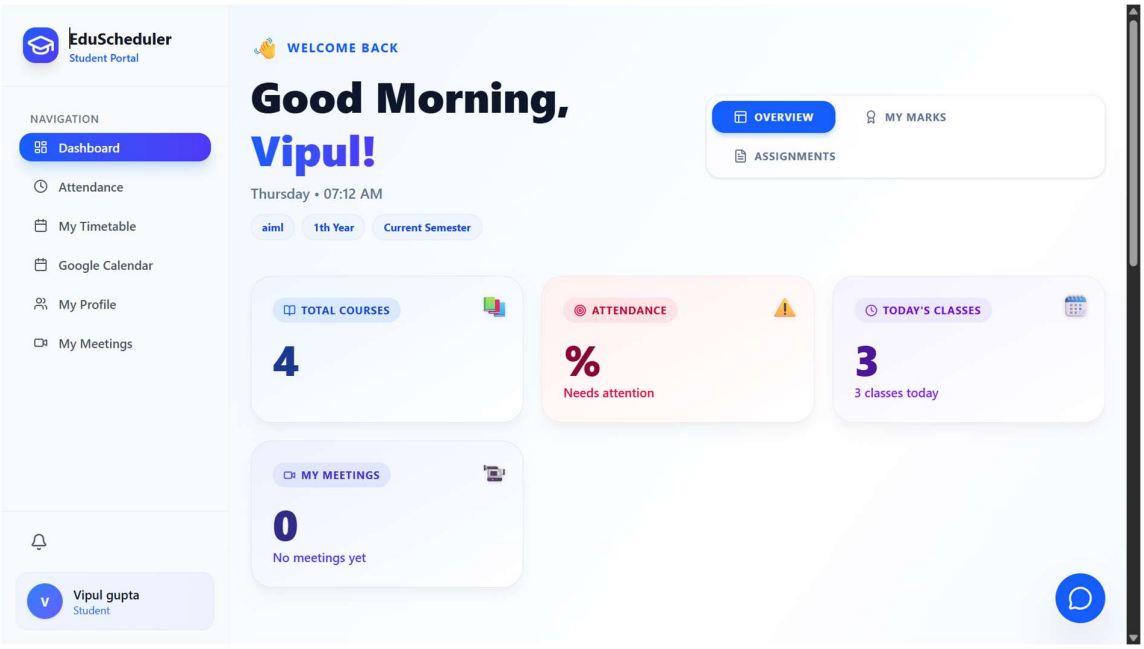

EduScheduler
Teacher Portal
NAVIGATION
Dashboard
Mark Attendance
My Schedule
Swap Classes
Apply Leave
My Requests
Room Management
Google Calendar
My Profile
My Meetings
pooja
Teacher
Good Evening, pooja!
Wednesday • 10:26 PM
electronics
ID: 8
Overview
Assessments
Assignments
Total Classes
0
Today's Classes
0
Pending Swaps
0
Leave Requests
0
My Meetings
0
Today's Schedule
On-Demand Broadcast
Instantly create a meeting and broadcast it to your students or colleagues.
Create Broadcast


EduScheduler
Teacher Portal
NAVIGATION
Dashboard
Mark Attendance
My Schedule
Swap Classes
Apply Leave
My Requests
Room Management
Google Calendar
My Profile
My Meetings
pooja
Teacher
Create Meeting Broadcast
Meeting Title *
e.g. Extra Class - Revisions
Description (Optional)
Brief agenda...
Timing
Start Now (Instant)
Schedule for Later
Target Audience *
Students
Teachers
Everyone
DEPARTMENT
Select Department...
YEAR
All Years
SECTION
All Sections
Meeting Link
On-Demand Broadcast
Instantly create a meeting and broadcast it to your students or colleagues.
Create Broadcast
Send Notification
Broadcast an announcement
Google Calendar Sync
Auto-sync your timetable as calendar events
Connected
synced
Last synced: 31/3/2026, 5:25:30 pm
Sync Now
Disconnect
Room Status Management
Weekly Overview


EduScheduler
Teacher Portal
Apply Leave
My Requests
Room Management
Google Calendar
My Profile
My Meetings
Send Notification
pooja
Teacher
Send to one individual user
Notification Title *
e.g., Unit 3 Notes Uploaded
Message *
Write the full announcement details here...
Attach File (optional · max 5MB)
Click to upload or drag & drop
PDF, Word (.docx), Excel (.xlsx), PNG, JPG · Max 5MB
Cancel
Send Notification
On-Demand Broadcast
Instantly create a meeting and broadcast it to your students or colleagues.
Create Broadcast


Assignment Management
Create Assignment
Published Assignments
No assignments created yet.


Google Calendar Sync
Auto-sync your timetable as calendar events
Connected
synced
Last synced: 31/3/2026, 5:25:30 pm
Sync Now
Disconnect
Request Class Swap
Select Class
-- Select a class --
Proposed Day
Monday
Proposed Time
09:00 AM
Reason
Cancel
Submit Request


EduScheduler
Student Portal
NAVIGATION
Dashboard
Attendance
My Timetable
Google Calendar
My Profile
My Meetings
Vipul gupta
Student
WELCOME BACK
Good Morning,
Vipul!
Thursday • 07:12 AM
aiml
1th Year
Current Semester
OVERVIEW
MY MARKS
ASSIGNMENTS
TOTAL COURSES
4
ATTENDANCE
%
Needs attention
TODAY'S CLASSES
3
3 classes today
MY MEETINGS
0
No meetings yet

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Additional Comments:

The developed system has evolved significantly from a basic timetable scheduler into a comprehensive Academic ERP integrated with communication and learning management capabilities. The project emphasizes real-world applicability, addressing key challenges faced by educational institutions such as scheduling conflicts, communication gaps, and academic data management.

Key enhancements such as true timetable swap logic, role-based meeting and notification systems, file-based content sharing, and bulk data management demonstrate a strong focus on usability and scalability. The integration of Google Calendar and automated meeting generation further bridges the gap between traditional academic systems and modern digital workflows.

The system is designed with a modular and extensible architecture, allowing future upgrades such as AI-based scheduling optimization, predictive performance analytics, and mobile application integration. Robust validation mechanisms and fallback strategies (e.g., manual link support) ensure system reliability even when external services are unavailable.

Overall, the project reflects a solid understanding of full-stack development, system design, database management, and real-world problem-solving, making it suitable for deployment in small to medium-scale academic environments and adaptable for further enterprise-level enhancements.

Signature of the Supervisor **Signature of the Project Lab Teacher**